# Development effects of electrification in Brazil

## Identification method and replication of main results of *Lipscomb et al (2013)*

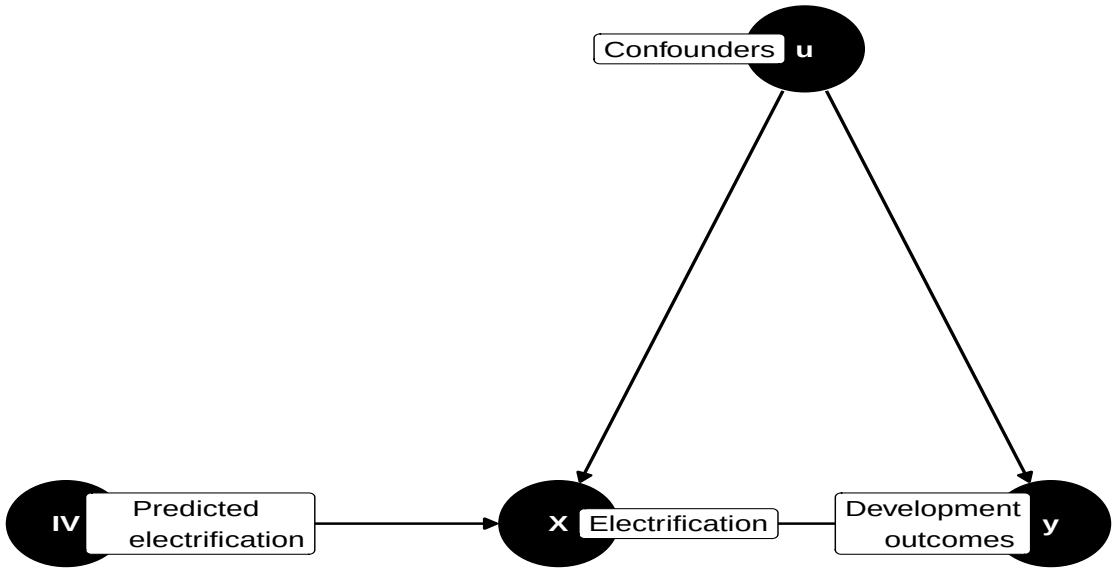
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# Throwback to the paper's research design

**Research question**: What's the effect of electrification on local economic development in Brazil?

- The paper relies on a natural experiment design by using an Instrumental Variable on the prediction of electricity availability.




▸ Ideal research design

# Identification strategy: fixed-effects IV (2SLS)

$$\textbf{First stage:} \quad E_{c,t-1} = \alpha_c^1 + \gamma_t^1 + \theta\, Z_{c,t-1} + \eta_{ct}$$

$$\textbf{Second stage:} \quad Y_{ct} = \alpha_c^2 + \gamma_t^2 + \beta\, \hat{E}_{c,t-1} + \varepsilon_{ct}$$

County $c$, decade $t$, 1960-2000. Electrification is lagged one decade.

**Causal identification if:**

**Relevance of the instrument**: met, first-stage $F \approx 25 > 10$.

**Exogeneity of the instrument**: no relationship between predicted electricity and confounders.

**Exclusion restriction**: geography affects local development only via electrification.

**Monotonicity**: not discussed but very plausible to hold.

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$E$: proportion of grid points in a county that are electrified in period $t$
$Z$: proportion of grid points in a county predicted to be electrified by the model
$\hat{E}$: instrumented electricity provision, predicted from the first stage

# Main findings and replication (I)

- **Housing values**: +10 pp electrification → **+881 reais**. It's a change of +6.8% of the mean value (Table 6).

| | Average value of housing | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Lagged electricity infrastructure | 5.023*** | 1.326*** | 0.8011*** | 8.468*** | 7.792*** | 8.811*** |
| | (0.8991) | (0.3007) | (0.2325) | (1.517) | (1.720) | (3.027) |
| $R^2$ | 0.15327 | 0.92211 | 0.92472 | 0.99955 | 0.99997 | 0.99997 |
| F-test (1st stage), Lagged electricity infrastructure | | | | 4,459.7 | 244.30 | 134.19 |
| Observations | 8,730 | 8,730 | 8,730 | 8,730 | 8,730 | 8,730 |
| Decade fixed effects | Yes | Yes | Yes | Yes | Yes | Yes |
| County fixed effects | | Yes | Yes | | Yes | Yes |
| Amazon x decade fixed effects | | | Yes | | | Yes |

**Note**: $R^2$ in the IV columns differs from the original due to differences in how R (fixest) and Stata compute $R^2$; coefficients are unaffected.

▸ Original Table 6

# Main findings and replication (II)

- **HDI**: moving from 0 to full electrification $\rightarrow$ **+0.11**. Would move the median county towards the 69th percentile county (Table 7).

| | Human development index | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Lagged electricity infrastructure | 0.0363** | 0.0091 | 0.0063 | 0.0908*** | 0.0912*** | 0.1093** |
| | (0.0141) | (0.0061) | (0.0044) | (0.0244) | (0.0292) | (0.0439) |
| $R^2$ | 0.65740 | 0.95956 | 0.96046 | 0.99988 | 0.99999 | 0.99999 |
| F-test (1st stage), Lagged electricity infrastructure | | | | 4,459.7 | 244.30 | 134.19 |
| Observations | 8,730 | 8,730 | 8,730 | 8,730 | 8,730 | 8,730 |
| Decade fixed effects | Yes | Yes | Yes | Yes | Yes | Yes |
| County fixed effects | | Yes | Yes | | Yes | Yes |
| Amazon x decade fixed effects | | | Yes | | | Yes |

**Note**: $R^2$ in the IV columns differs from the original due to differences in how R (fixest) and Stata compute $R^2$; coefficients are unaffected.

▸ Original Table 7

# Are the results convincing?

## Convincing

- IV uses cost-driven variation in grid expansion, removing demand-driven placement (endogeneity).
- Results are robust after controls for the geographic factors used to construct the IV.

## Not convincing

**Omitted variables**: grid may come with package of infrastructure. Only partially addressed as roads are only proxied.

**Mechanisms**: migration data only for 1990-2000, while most dams were built earlier $\rightarrow$ possible missing variable between X and Y.

**Dam effects**: changes in local hydrology might affect local economic development (flood control).

**Measurement**: all dams are considered the same size and with a fixed 50 km reach. HDI is the only welfare measure.

# Appendix: Research design: ideal benchmark vs. the paper

## 1. RCT: randomized grid rollout

Among areas where grid expansion is feasible, a lottery decides which get the grid now and which wait ten years.

## 2. Staggered diff-in-diff

Valid if construction timing depended on administrative factors unrelated to local growth (parallel trends). Here assignment is not random so is not plausible.

## 2b. Instrumented diff-in-diff (TBD)

Combine staggered timing with an instrument for when areas are electrified, to deal with endogenous placement. Possible avenue to research.

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# Appendix: Original table 6

TABLE 6—HOUSING VALUES
DEPENDENT VARIABLE: AVERAGE VALUE OF HOUSING

| | OLS | | | IV | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Lagged electricity infrastructure | 5.023*** | 1.326*** | 0.801*** | 8.468*** | 7.792*** | 8.811*** |
| | (0.90) | (0.35) | (0.27) | (1.52) | (1.72) | (3.03) |
| Year FE? | Yes | Yes | Yes | Yes | Yes | Yes |
| County FE? | No | Yes | Yes | No | Yes | Yes |
| Amazon × year dummies[a] | No | No | Yes | No | No | Yes |
| $R^2$ | 0.153 | 0.922 | 0.925 | 0.106 | 0.191 | 0.151 |
| Observations | 8,730 | 8,730 | 8,730 | 8,730 | 8,730 | 8,730 |

*Notes:* Standard errors clustered by county in parentheses. The dependent variable is average housing value in thousands of reais. All regressions have county size weights and year dummies. The average housing value in the sample is 13.048.

[a]Topographic factor is interacted with a full set of decade fixed effects in order to flexibly control for differential trends by that Topographic factor.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

# Appendix: Original table 7

TABLE 7—HUMAN DEVELOPMENT INDEX

| | OLS | | | IV | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Lagged electricity infrastructure | 0.036** | 0.009 | 0.006 | 0.091*** | 0.091*** | 0.109** |
| | (0.01) | (0.01) | (0.01) | (0.02) | (0.03) | (0.04) |
| Year FE? | Yes | Yes | Yes | Yes | Yes | Yes |
| County FE? | No | Yes | Yes | No | Yes | Yes |
| Jungle × year dummies[a] | No | No | Yes | No | No | Yes |
| $R^2$ | 0.657 | 0.960 | 0.960 | 0.640 | 0.931 | 0.930 |
| Observations | 8,730 | 8,730 | 8,730 | 8,730 | 8,730 | 8,730 |

*Notes:* Standard errors clustered by county in parentheses. The dependent variable is the human development index. Year dummies are included in all regressions. All regressions have county size weights. The average HDI value in the sample is 0.557.

[a]Topographic factor is interacted with a full set of decade fixed effects in order to flexibly control for differential trends by that Topographic factor.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.